

Muhammad Rehan Khalid

AI Engineer specializing in LLM systems, Agentic AI, and scalable production ML platforms

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I have a valid work permit for Germany. My notice period is 30 days.



TECHNICAL SKILLS

AI Engineer (5+ years of project experience) **Programming Languages:** Python **AI / Machine Learning:** Pytorch | Machine Learning | Deep Learning | NLP | LLMs | Huggingface | scikit-learn | RAG | Transformers | Prompt Engineering **LLM & Agent Frameworks:** LangGraph | LlamaIndex | LangChain | OpenAI API | DeepEval | RAGAS | Hugging Face Transformers **Backend & APIs:** FastAPI | REST APIs **Data & Databases:** Redis | Pyspark | PostgreSQL | Vector Databases | Weaviate **Cloud Platforms:** AWS | Azure **DevOps & MLOps:** Docker | Kubernetes | CI/CD | MLflow | Kubeflow **Streaming & Messaging:** Kafka **Version Control & Collaboration:** Git | Agile

WORK EXPERIENCE

07/2025 - today

Data Science Intern - LLM

CHECK24 Vergleichsportal GmbH, München, Germany

Project: DocGPT Document RAG Platform - AI-based document extraction and chat system used by CHECK24 verticals

- Built a **graph-based orchestration layer** for agentic document processing, enabling multi-step workflows (classification → targeted extraction) and flexible chaining of extraction use cases within a unified system
- Implemented a Retrieval-Augmented Generation module for the DocGPT extraction and chat platform, enabling long-document support beyond LLM context limits and onboarding additional CHECK24 verticals.
- Built multi-format ingestion and indexing for PDF, DOCX, PPT, web and Notion sources with cleaning, deduplication, metadata tagging and chunking; stored embeddings in Weaviate for fast semantic retrieval.
- Developed hybrid retrieval workflows with BM25/vector search, re-ranking and query rewriting, improving grounded Q&A and structured invoice/contract extraction into normalized JSON.
- Enabled onboarding of CHECK24 Autoteile (Reifen) by building an agentic scan-preprocessing pipeline (dewarp, deblur, lighting/shadow correction), improving OCR/LLM input quality and supporting ~1K API calls per day in production.
- Developed a real-time monitoring dashboard for request volume, latency and error rates, reducing 5xx errors by about 15% and helping scale DocGPT extraction usage to 80K+ weekly API requests.

Technologies used: Python | LLMs | RAG | Weaviate | Langchain | FastAPI | AWS | Docker | CI/CD

12/2024 - 06/2025

Research Assistant - LLM

Fraunhofer IAIS, Sankt Augustin, Germany

Project: German Legal AI Assistant: NLP-driven legal QA platform trained on 30K+ documents for internal enterprise use

- Fine-tuned LLMs (LLaMA-3, Gemma) using LoRA on 23,905 QA pairs, improving model accuracy from 36.8% to 56.4%
- evaluated and optimized response quality using LLM-as-a-judge, achieving a 50% increase in response relevance.
- Designed a retrieval-augmented fine-tuning (RAFT) approach, improving factual correctness by 20% and significantly reducing hallucinations
- Built a multi-step RAG pipeline with advanced retrieval and re-ranking, boosting answer precision by 25% and context recall by 30%

01/2024 - 11/2024	Technologies used: Python LLMs PEFT LoRA Unsloth PyTorch RAGAS DeepEval RAG Research Assistant - MLOps Fraunhofer IVI, Dresden, Germany Project: Autonomous Driving Signal Monitoring Platform - Germany-based geospatial MLOps system for AD-drop and RSU status monitoring - Built Grafana dashboards for Autonomous Driving signal quality and AD-drop monitoring using OpenStreetMap data, enabling real-time outage alerts. - Enriched TimescaleDB datasets with Roadside Unit and bus-stop metadata, reducing query latency by 40%. - Managed and optimized TimescaleDB data workflows, implementing retention policies and integrity constraints to improve reliability. Technologies used: Python Grafana TimescaleDB PostgreSQL OpenStreetMap
12/2021 - 08/2023	Data Analyst i2c Inc., Lahore, Pakistan Project: AI Fraud Detection Platform - USA-based fraud detection engine for real-time transaction monitoring used by financial institutions - Led fraud detection projects using ensemble models (XGBoost, LightGBM) and neural networks (TabNet, RNNs), capturing 40% more fraud at 1:5 false-positive odds. - Optimized feature pipelines, reducing processing time from 5 hours to 10 minutes and improving data freshness by 95% - Implemented real-time fraud prevention models, reducing annual losses by \$150K and improving detection response for high-risk transactions. - Engineered profiling pipelines generating rolling-window features from minutes to 30 days and reduced class imbalance from 1/1000 to 1/100 through data sampling approach. Technologies used: Python XGBoost LightGBM TabNet Feature Engineering PostgreSQL
08/2020 - 12/2021	Data Scientist Afiniti, Lahore, Pakistan Project: Customer Propensity Modeling Platform - customer ranking and experimentation engine for revenue optimization - Built LightGBM-based customer propensity models to rank high-value customers and support targeted revenue optimization decisions, driving a 10% revenue lift. - Designed propensity scoring features from customer interaction and behavioral data, enabling more precise segmentation of high-potential customers. - Ran A/B tests and ROI analyses contributing to a +6% client-side sales uplift. Technologies used: Python LightGBM Machine Learning Customer Propensity Modeling A/B Testing

EDUCATION

10/2024 - 04/2026	MSc Applied Data Science George August Universität, Göttingen, Germany
10/2016 - 08/2020	BS Math and Economics Lahore University of Management and Science, Lahore, Pakistan

LANGUAGE SKILLS

English (Mother tongue) | German (B1, currently learning)

HOBBIES

Squash | Reading | Baking